Abstract Algebra - Homework 2

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Problem 1

Let $|G| = p^\alpha m$, where p is a prime, $\alpha \geq 1$ and $p \nmid m$. By definition of a p-Sylow subgroup, we have $|P| = p^\alpha$. Since $N \lhd G$, Lagrange's theorem implies $|N| \mid |G|$. Thus, we can write:

$$|N| = p^\beta d,$$

with $\beta \in [1, \alpha]$ and $d \mid m$, where $p \nmid d$. Consider the subgroup $N \cap P$. Since $N \cap P \leq P$, again applying Lagrange's theorem gives $|N \cap P| \mid |P| = p^\alpha$. Therefore, we must have:

$$|N \cap P| = p^\gamma,$$

where $\gamma \leq \beta$, hence $N \cap P$ must be a p-subgroup of N.

Consider the subgroup NP which contains N and P. Since $N \lhd G$ and $P \leq G$ we may apply the following formula,

$$|NP| = \frac{|N||P|}{|N \cap P|} = \frac{p^{\alpha+\beta}d}{p^\gamma} = p^{\alpha+\beta-\gamma}d$$

Since P is a p-Sylow subgroup of G it must also be a p-Sylow subgroup in any subgroup of G containing it. Therefore, its index in NP must be relatively prime to p.

$$[NP : P] = \frac{|NP|}{|P|} = \frac{p^{\alpha+\beta-\gamma}d}{p^\alpha} = p^{\beta-\gamma}d \implies p^{\beta-\gamma} = 1 \implies \beta = \gamma.$$

This shows $|N \cap P| = p^\beta$, which means it is a p-Sylow subgroup of N.

The quotient group G/N inherits a finite order, $|G/N| = \frac{|G|}{|N|} = \frac{p^\alpha m}{p^\beta d} = p^{\alpha-\beta}\frac{m}{d}$. A p-Sylow subgroup must therefore be of order $p^{\alpha-\beta}$. Since we concluded $\gamma = \beta$, $|NP| = p^\alpha d$. Moreover $|N| = p^\beta d$. This allows us to construct a candidate subgroup with the correct order, NP/N

$$|NP/N| = \frac{|NP|}{|N|} = \frac{p^\alpha d}{p^\beta d} = p^{\alpha-\beta}$$

Finally, since NP is a subgroup of G containing N, the quotient NP/N is a subgroup of G/N. Therefore, NP/N is a subgroup of order $p^{\alpha-\beta}$ in G/N, and hence is a p-Sylow subgroup of G/N.

Problem 2

(1) Naturally, since both P and Q are Sylow p-subgroups of the finite group G, the well-known Sylow theorem assures us P and Q are conjugate.

(2) **($\subseteq$)** Let $g \in N_G(P)$, so by definition $gPg^{-1} = P$. Since f is a homomorphism, we have:

$$f(g)f(P)f(g)^{-1} = f(gPg^{-1}) = f(P),$$

which implies $f(g) \in N_H(f(P))$. Therefore, $f(N_G(P)) \subseteq N_H(f(P))$.

($\supseteq$) Now let $h \in N_H(f(P))$. Since f is surjective, there exists $g \in G$ such that $f(g) = h$. Consider the conjugate subgroup $gPg^{-1} \leq G$. Then:

$$f(gPg^{-1}) = f(g)f(P)f(g)^{-1} = hf(P)h^{-1} = f(P),$$

so $f(gPg^{-1}) = f(P)$. By previous section, it follows that gPg^{-1} and P are conjugate in G; that is, there exists $x \in G$ such that $xgPg^{-1}x^{-1} = P$. Define $y = xg$, then:

$$yPy^{-1} = xgPg^{-1}x^{-1} = P \Rightarrow y \in N_G(P),$$

and since $f(y) = f(xg) = f(x)f(g)$, we get:

$$f(g) = f(x)^{-1}f(y) \in f(N_G(P)).$$

Thus $h = f(g) \in f(N_G(P))$, so $N_H(f(P)) \subseteq f(N_G(P))$.

(3) Since $P \leq G$ and f is a group homomorphism, we have $f(P) \leq H$, so $|f(P)| \mid |H|$. By the First Isomorphism Theorem, we have:

$$P/\ker(f|_P) \cong f(P),$$

so

$$|f(P)| = \frac{|P|}{|\ker(f|_P)|} = \frac{p^\alpha}{|\ker(f|_P)|},$$

which implies that $|f(P)| = p^\beta$ for some $\beta \in [0, \alpha]$.

Since $p \nmid |H|$ and $|f(P)| \mid |H|$, the only possibility is $|f(P)| = 1$. Therefore,

$$f(P) = \{e\} \subseteq \ker(f).$$

From previous section, we know that $f(N_G(P)) = N_H(f(P))$. Since $f(P)$ is trivial, its normaliser is all of H, so:

$$N_H(f(P)) = H \Rightarrow f(N_G(P)) = H.$$

Problem 3

(1) (a) By Sylow's theorems, the number n_5 of 5-Sylow subgroups satisfies $n_5 \equiv 1 \mod 5$ and divides 3, so $n_5 = 1$. A unique Sylow subgroup is always normal, so G has a normal subgroup of order 5.

(b) Let P_3 and P_5 be the Sylow 3- and 5-subgroups of G, with $|P_3| = 3$ and $|P_5| = 5$. Since $|P_3|$ and $|P_5|$ are relatively prime and both subgroups are cyclic, as all groups of prime order are, and P_5 is normal, it follows that:

$$G \cong P_3 \times P_5 \cong \mathbb{Z}/3 \times \mathbb{Z}/5 \cong \mathbb{Z}/15,$$

so G is cyclic.

(2) Again by Sylow's theorems, the number n_p of p-Sylow subgroups satisfies $n_p \equiv 1 \mod p$ and divides q, so $n_p = 1$ or q. Similarly, $n_q \equiv 1 \mod q$ and divides p, so $n_q = 1$ or p. Suppose for contradiction that $n_q = p$. Then:

$$p \equiv 1 \mod q \Rightarrow p \equiv 1 \pmod{q},$$

which contradicts the assumption $p \not\equiv 1 \pmod q$. Therefore, $n_q = 1$, and the q-Sylow subgroup is unique and hence normal.

All groups of prime order are cyclic, so both the p-Sylow and q-Sylow subgroups are cyclic. Let P and Q denote the p- and q-Sylow subgroups, with $Q \trianglelefteq G$ and $P \cap Q = \{e\}$. Since $|P||Q| = pq = |G|$ and Q is normal, we have:

$$G \cong P \times Q \cong \mathbb{Z}/p \times \mathbb{Z}/q \cong \mathbb{Z}/pq,$$

because p and q are relatively prime. Hence, G is cyclic.

(3) (a) Once again by Sylow's theorems, the number n_3 of 3-Sylow subgroups satisfies $n_3 \equiv 1 \mod 3$ and divides 4. Thus, $n_3 = 1$ or 4. The number n_2 of 2-Sylow subgroups satisfies $n_2 \equiv 1 \mod 2$ and divides 3. So $n_2 = 1$ or 3.

If either $n_3 = 1$ or $n_2 = 1$, the corresponding Sylow subgroup is unique and hence normal. Therefore, G always has a normal subgroup.

(b) An example of a non-cyclic group of order 12 is the alternating group A_4, the group of even permutations on 4 elements. It has order 12 but is not cyclic.

Problem 4

Both a and b are products of two disjoint 4-cycles, so $|a| = |b| = \mathrm{lcm}(4, 4) = 4$. Since a and b commute, i.e they act on disjoint or compatibly permuted sets, H is abelian.

We compute that $|H| = 8$, and H is generated by two commuting elements of order 4. Therefore, H is a finite abelian group of order 8.

By the classification of finite abelian groups, the only abelian groups of order 8 are:

$$\mathbb{Z}_8, \quad \mathbb{Z}_4 \times \mathbb{Z}_2, \quad \mathbb{Z}_2 \times \mathbb{Z}_2 \times \mathbb{Z}_2.$$

Since H contains elements of order 4 but none of order 8, it cannot be cyclic, so:

$$H \cong \mathbb{Z}_4 \times \mathbb{Z}_2.$$

Problem 5

(1) Assume G/Z is cyclic, so there exists $g \in G$ such that every element of G/Z is of the form $g^n Z$ for some $n \in \mathbb{Z}$. Then for any $x \in G$, there exists $n \in \mathbb{Z}$ and $z \in Z$ such that:

$$x = g^n z.$$

Now let $x = g^n z$ and $y = g^m z'$, with $z, z' \in Z$. Then:

$$xy = g^n z \cdot g^m z' = g^{n+m} z z', \quad \text{and} \quad yx = g^m z' \cdot g^n z = g^{m+n} z' z.$$

Since $Z \subseteq Z(G)$, we have $zz' = z'z$, so:

$$xy = yx.$$

Thus, any two elements of G commute, so G is abelian.

(2) Since G is a p-group, its center $Z(G)$ is nontrivial. Hence, $|Z(G)| = p$ or p^2. If $|Z(G)| = p^2$, then $Z(G) = G$, so G is abelian. If $|Z(G)| = p$, then the quotient group $G/Z(G)$ has order p and is cyclic. By previous section, if $G/Z(G)$ is cyclic and $Z(G)$ is central, then G is abelian. Therefore, in both cases, G is abelian.

(3) Let p be a prime. Up to isomorphism, there are exactly two groups of order p^2:

$$\mathbb{Z}_{p^2}, \quad \mathbb{Z}_p \times \mathbb{Z}_p.$$

These are the only abelian groups of order p^2, and by previous sectio, all groups of this order are abelian. Therefore, this list is complete up to isomorphism.

Problem 6

(1) Since $A \cong \mathbb{Z}_n$, any homomorphism $\chi \in \widehat{A}$ is determined by the image of a generator, and this image must be an n-th root of unity. Therefore, $\widehat{A} \cong \mu_n$, the group of n-th roots of unity, which is cyclic of order n. Hence, $\widehat{A}$ is cyclic of order n.

(2) By the structure theorem for finite abelian groups, A is isomorphic to a product of cyclic groups:

$$A \cong \mathbb{Z}_{n_1} \times \cdots \times \mathbb{Z}_{n_k}.$$

Then,

$$\widehat{A} \cong \widehat{\mathbb{Z}_{n_1}} \times \cdots \times \widehat{\mathbb{Z}_{n_k}},$$

and each $\widehat{\mathbb{Z}_{n_i}} \cong \mathbb{Z}_{n_i}$. Therefore, $\widehat{A} \cong A$, so $|\widehat{A}| = |A| = n$.